

IDENTIFICATION

CODE : GI-4-S1-EC-ISD
ECTS : 2

HOURS

Cours : 0h
TD : 18h
TP : 16h
Projet : 0h
Evaluation : 2h
Face à face pédagogique : 36h
Travail personnel : 12h
Total : 48h

ASSESSMENT METHOD

TEACHING AIDS

Course handouts

TEACHING LANGUAGE

English

CONTACT

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AIMS

This course belongs to teaching unit Gestion et exploitation des données de l'entreprise (GI-4-S1-UE-GEDE) and contributes to the following skills :

A3 Implement an experimental approach (Level 2)
A5 Process data (Level 3)
A6 Report on an analysis or on a scientific approach (Level 3)
C1 Observing, measuring, analyzing and interpreting an activity or a system from data (Level 3)
C4 Sizing the hardware and / or software of a system (Level 2)
C11 Understanding and evaluating a structure holistically, through socio-economic frames of reference (Level 1)
B2 work, learn, progress autonomously (Level 2)
B3 interact with people and work in a team (Level 3)

By allowing the engineering students to work and be evaluated on the following skills:

- Mastering structured data processing tools: Excel and Stats (C4)
- Know how to make a decision based on numerical results (C1)
- Choose a processing method based on the data and objectives (A3, A5, A6, C1, C11, B2, B3)

CONTENT

DATA ANALYSIS

1.Data

Characteristics: origin, quality, coding

Preparation: restoration, filtering, synchronization, construction of classes

Dependence: distances, ultra metric, correlation

2.Tools :

Descriptive methods: principal components analysis, correspondences, preferences

Explanatory methods: regression, segmentation, canonical analysis, discriminated analysis

Structural methods: ultra metric and hierarchical classification, dynamic swarms.

3.Analysis :

Analysis methodologies

Interpretation rules

Case studies

BIBLIOGRAPHY

**** Data analysis ****

1. E.Diday, J.Lemaire, J.Pouget, F.Testu, Eléments d'analyse de données, Dunod 1982
2. G.Saporta, Probabilités, Analyse de données et Statistiques, Technip 1990
3. Jean-Pierre , Analyse interactive des données (ACP, AFC) avec Excel 2000 : théorie et pratique / Georgin. Presses universitaires de Rennes , 2002
4. Alain Morineau et Yves-Marie Chatelin , L'analyse statistique des données [Texte imprimé] : apprendre, comprendre et réaliser avec Excel, Ellipses , 2005 , 407 p

**** Data warehouse ****

1. Entrepôts de données. Guide pratique de modélisation dimensionnelle, 2ème édition, R. Kimball, M. Ross
2. Piloter l'entreprise grâce au data Warehouse, J.-M. Franco, S de Lignerolles

PRE-REQUISITES

Statistics, data base